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CSA0707

SIMATS ENGINEERING
ASSESSMENT TOOL 2

38

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration 2	Limited participation	No collaboration or contribution 2
		9	10	9	10

CO4-AT2

Star topology is a suitable because all devices are connected to a central switch.

* A 24 port managed switch can connect computers, printers & VoIP phones with extra ports for future use.

* UTP cables can be used to provide high speed connections with the office.

* A router / firewall connects the office network to the Internet & provides security & DHCP services.

* A wifi access point can provide wireless connectivity for laptops & mobile devices.

* If one computers (or) cable fails, only that is affected, making the network easy to manage & troubleshoot.

* Bank network - Mesh Topology

* A full (or) partial mesh topology is suitable for a bank because high reliability is required

* Device have multiple connections & alternative paths for data transmissions

* If one link fails, traffic can automatically use another path.
* Reductant routers, switches & firewalls can be used to avoid a single point of failure.
* fiber-optic cables provide high speed, low latency & resistance to electromagnetic interface.
* mesh topology provides high fault tolerance and near-continuous operation, making it suitable for critical banking network.

3. Small classroom lab - Bus topology

* Bus topology is suitable for a small classroom with a limited budget.
* Traditional bus network can use cable with connectors & terminators.
* It is the simple & inexpensive to install for a small numbers of computers.
* The main disadvantages is that a break in the backbone cable can affect the entire network.

... is suitable for light network usage, such as browsing, filesharing, & basic classroom activity.

4. A hybrid topology is best suited for a large university campus because a single topology cannot efficiently handle thousands of users across multiple buildings.

- Campus backbone: Use a fiber-optic ring or partial mesh topology with high-speed 10/40 Gbps links. This provides redundancy, so a single link failure does not disconnect a building.

- Building level: use a star topology connects computers & devices to switches.

- Labs: Bus/tree topology can be used to reduce cost.

- Wi-fi: Access points connect to switches using PoE.

- Core → Distribution → Access: This makes the network easy to manage & expand.

output :-

5. Computing with multiple departments.

* Each department such as HR, finance, salary & IT. Can have its own network segment
* Layer 2 switches connect computers, printers & other devices in each department.
* VLAN's are used to logically separate different department & improve security.
* Layer 3 switch (or) router enables communica b/w different VLANS.
* A router / firewall connects the internet
* network to the Internet & provides additional security.
* A Subnetting, redundant links, & proper I
* Addressing improves perdolming reliabilit & future expansion.